

SYDNEY TECHNICAL HIGH SCHOOL

MATHEMATICS EXTENSION 1

YEAR 11 YEARLY EXAMINATION

SEPTEMBER 2003

Time allowed: 90 minutes

Instructions:

- Show all necessary working
- Start each question on a new page
- Marks will not be awarded for careless or badly arranged work
- Non-programmable calculators may be used
- This paper must be handed in with your answer sheets
- Answers must be written in blue or black pen

Name: _____

Teacher: _____

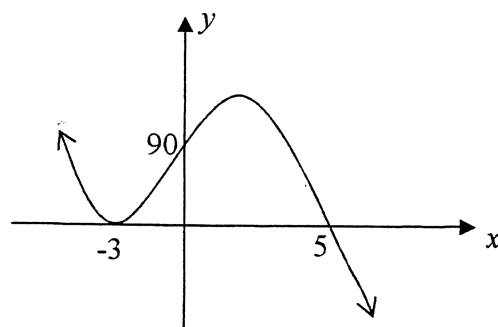
Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total
/11	/10	/10	/10	/10	/12	/63

Question 1 (11 marks)

- a) If $f(n) = n(n+1)(n+2)$
- i) Simplify $\frac{f(n)}{f(n+1)}$ (1)
- ii) Express $f(n) - f(n+1)$ in factored form (2)
- b) i) Find the gradient of the normal to the curve
 $y = x^4 + x^{3/2}$ at A (1, 5) (2)
- ii) Find the acute angle to the nearest degree between the normal and the line $2x + 3y = 7$ (2)
- c) If α, β and γ are the roots of
 $2x^3 + 12x^2 - 6x + 1 = 0$ find
- i) $\alpha + \beta + \gamma$ (1)
- ii) $\alpha\beta\gamma$ (1)
- iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ (2)

Question 2 (10 marks)**Start a new page**

- a) Find the equation of the cubic polynomial in the sketch below (2)



- b) Solve $\frac{1}{x-1} - \frac{1}{x} > 0$ (3)
- c) Find the equation of the parabola with vertex (2, -1), axis of symmetry parallel to the y axis and passing through the point (8, 2) (2)
- d) Prove that $3kx^2 - (2k + 3n)x + 2n = 0$ has rational roots if k and n are rational (3)

Question 3 (10 marks) Start a new page

a) Solve $2 \sin x - 3 \cos x = 2$ for $0^\circ \leq x \leq 360^\circ$ (3)

b) Prove $\frac{1 - \cos 2x}{\sin 2x} = \tan x$ (2)

Hence find the exact value of $\tan 15^\circ$ in simplest form (2)

c) The monic polynomial $P(x)$ has a degree of 2. When $P(x)$ is divided by x the remainder is -6. If $P(3) = P(-5)$, find the polynomial. (3)

Question 4 (10 marks) Start a new page

a)

i) The polynomial equation $P(x) = 0$ has a double root at $x = a$. By putting $P(x) = (x - a)^2 Q(x)$, show that $P'(a) = 0$ (2)

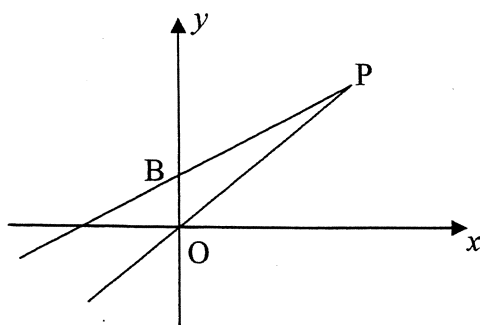
ii) You are told that the equation $mx^4 + nx^3 - 6x^2 + 22x - 12 = 0$ has a double root at $x = 1$. Find the values of m and n . (3)

b) Find the values of m for which the line $y = mx + 4$ intersects the curve $y = x^2 - 4x + 5$ at two distinct points (3)

c) Find the co-ordinates of the point that divides the interval joining $A(3,2)$ to $B(-1,1)$ externally in the ratio 3:2. (2)

Question 5 (10 marks) Start a new page

a)

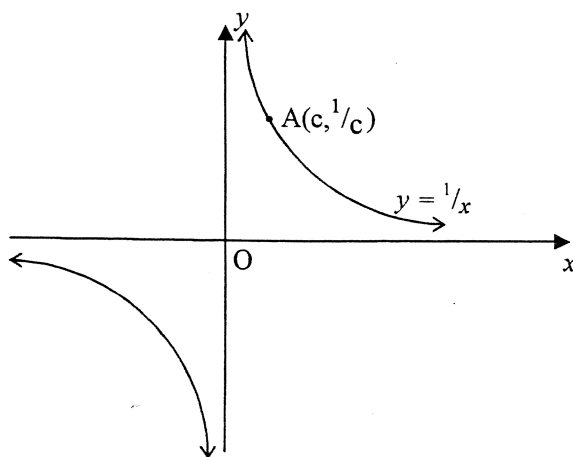


B is the point $(0, 3)$. The gradient of BP is m and the gradient of OP is $2m$.

O is the origin.

- i) Write the equations of the lines BP and OP (2)
- ii) Find the co-ordinates of P (1)
- iii) Find the locus of P as the value of m varies. (1)

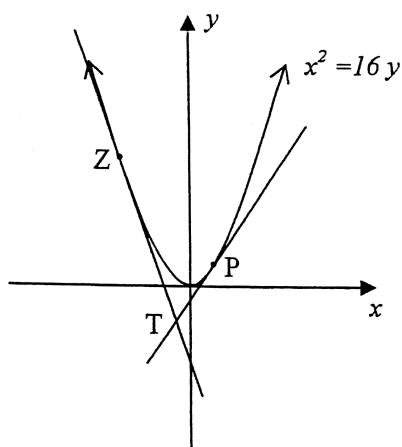
b) The point $A(c, \frac{1}{c})$ lies on the curve $y = \frac{1}{x}$



- i) Find the equation of the tangent at A (2)
- ii) The tangent at A cuts the x axis at B and the y axis at C. Find the coordinates of B and C. (2)
- iii) Show that the area of triangle BOC is a constant (2)

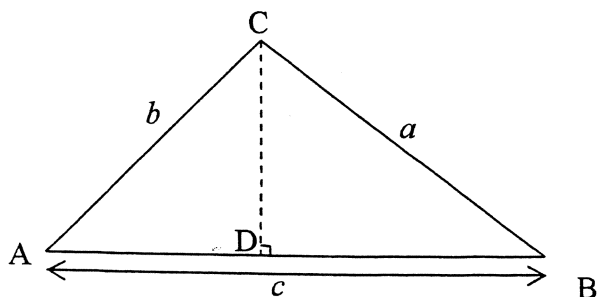
Question 6 (12 marks) Start a new page

a)



- i) Find the equation of the tangent to the curve $x^2 = 16y$ at $P(8p, 4p^2)$ (2)
- ii) If this tangent at P also passes through T $(-2, -2)$ and TZ is another tangent to the parabola at Z, find the coordinates of P and Z (2)
- iii) Let RQ be a chord on the same parabola with $R(8r, 4r^2)$ and $Q(8q, 4q^2)$. Prove that the equation of RQ is $y = \left(\frac{r+q}{2}\right)x - 4rq$ (2)
- iv) If RQ is a focal chord show that $rq = -1$ (1)

b)



The triangle ABC has sides of length a, b and c as shown in the diagram.

The point D lies on AB and CD is perpendicular to AB

- i) Show that $a \sin B = b \sin A$ (1)
- ii) Show that $c = a \cos B + b \cos A$ (1)
- iii) Hence given that $c^2 = 4ab \cos A \cos B$ show that $a = b$ (3)

End of Paper.

Question 1

$$a) i) \frac{f(n)}{f(n+1)} = \frac{n(n+1)(n+2)}{(n+1)(n+2)(n+3)}$$

$$= \frac{n}{n+3}$$

$$ii) f(n) - f(n+1)$$

$$= n(n+1)(n+2) - (n+1)(n+2)(n+3)$$

$$= (n+1)(n+2)(n - (n+3))$$

$$= \underline{\underline{-3(n+1)(n+2)}}$$

$$b) i) y = x^4 + x^{3/2}$$

$$\frac{dy}{dx} = 4x^3 + \frac{3}{2}x^{1/2}$$

$$\text{at } A(1, 5) \quad m_T = \frac{11}{2} \quad \therefore m_{\text{normal}} = -\frac{2}{11}$$

$$ii) \tan \theta = \left| \frac{-\frac{2}{11} - (-\frac{2}{3})}{1 + (-\frac{2}{11} \times -\frac{2}{3})} \right|$$

$$(m_1 = -\frac{2}{11} \quad m_2 = -\frac{2}{3})$$

$$\tan \theta = \left| \frac{16/33}{37/33} \right|$$

$$\theta = \underline{\underline{23^\circ}}$$

$$c) a = 2 \quad b = 12 \quad c = -6 \quad d = 1$$

$$i) \alpha + \beta + \gamma = \underline{\underline{-\frac{12}{2} = -6}}$$

$$ii) \angle \beta \gamma = \underline{\underline{-\frac{1}{2}}}$$

$$iii) \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

$$\frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{-6/2}{-1/2}$$

$$= \underline{\underline{6}}$$

Question 2

$$a) P(x) = A(x+3)^2(x-5)$$

$$\text{sub } x=0 \quad P(0) = 90$$

$$90 = A \cdot 3^2 \cdot x - 5$$

$$90 = -45A$$

$$\therefore A = -2$$

$$\underline{\underline{P(x) = -2(x+3)^2(x-5)}}$$

$$b) \frac{1}{x-1} - \frac{1}{x} > 0$$

$$\frac{x - (x-1)}{x(x-1)} > 0$$

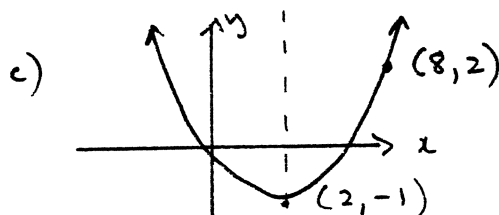
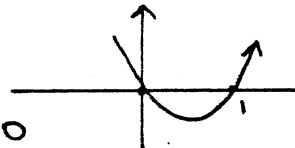
$$\frac{1}{x(x-1)} > 0$$

$$\frac{x^2(x-1)^2}{x(x-1)} > 0$$

$$x(x-1)$$

$$x(x-1) > 0$$

$$\underline{\underline{x > 1, \quad x < 0}}$$



$$(x-2)^2 = 4a(y+1) \quad \text{sub } (8, 2)$$

$$6^2 = 4a \cdot 3$$

$$\therefore a = 3$$

$$\therefore \text{eqn } \underline{\underline{(x-2)^2 = 12(y+1)}}$$

d) Roots rational if Δ square

$$\Delta = (2k+3n)^2 - 4 \cdot 3k \cdot 2n$$

$$= 4k^2 + 12kn + 9n^2 - 24kn$$

$$= 4k^2 - 12kn + 9n^2$$

$$\underline{\underline{\Delta = (2k-3n)^2 \quad \text{if } k, n \text{ rational}}}$$

Question 3

$$a) R = \sqrt{4+9} = \sqrt{13}$$

$$2\sin x - 3\cos x = \sqrt{13} \left[\frac{2}{\sqrt{13}} \sin x - \frac{3}{\sqrt{13}} \cos x \right]$$

place in form $\sqrt{13}(\sin(x-\alpha))$
 i.e. $\sqrt{13}[\sin x \cos \alpha - \cos x \sin \alpha]$

$$\therefore \cos \alpha = \frac{2}{\sqrt{13}} \quad \sin \alpha = \frac{3}{\sqrt{13}}$$

α is acute $\alpha = 56.31^\circ$
 or $56^\circ 19'$

$$\sqrt{13} \sin(x - 56^\circ 19') = 2$$

$$\sin(x - 56^\circ 19') = \frac{2}{\sqrt{13}} \quad \begin{array}{c} \checkmark \\ S \\ T \end{array} \begin{array}{c} A \\ C \end{array} \checkmark$$

$$x - 56^\circ 19' = 33^\circ 41', 146^\circ 19'$$

$$\therefore x = 90^\circ, 202^\circ 38'$$

$$b) LHS = \frac{1 - \cos 2x}{\sin 2x}$$

$$= \frac{1 - [\cos^2 x - \sin^2 x]}{2 \sin x \cdot \cos x}$$

$$= \frac{2 \sin^2 x}{2 \sin x \cos x}$$

$$= \frac{\sin x}{\cos x}$$

$$= \tan x$$

$$= RHS$$

$$\therefore \tan 15^\circ = \frac{1 - \cos 30^\circ}{\sin 30^\circ}$$

$$= \left(1 - \frac{\sqrt{3}}{2}\right) \div \frac{1}{2}$$

$$= \frac{2 - \sqrt{3}}{2} \times \frac{2}{1}$$

$$= \underline{\underline{2 - \sqrt{3}}}$$

$$c) P(x) = x^2 + bx + c$$

$$P(0) = -6 \quad \therefore c = -6$$

$$P(3) = 9 + 3b - 6 = 3 + 3b$$

$$P(-5) = 25 - 5b - 6 = 19 - 5b$$

$$\therefore 3 + 3b = 19 - 5b$$

$$8b = 16$$

$$b = 2$$

$$\therefore P(x) = x^2 + 2x - 6$$

Question 4

$$a) i) P(x) = (x-a)^2 Q(x)$$

$$P'(x) = 2(x-a) \cdot Q(x) + (x-a)^2 Q'(x)$$

$$P'(a) = 2(a-a)Q(a) + (a-a)^2 Q'(a)$$

$$\therefore P'(a) = 0$$

$$ii) \text{ double root } a + x = 1$$

$$\therefore P(1) = 0 \quad P'(1) = 0$$

$$P(x) = mx^4 + nx^3 - 6x^2 + 22x - 12$$

$$P'(x) = 4mx^3 + 3nx^2 - 12x + 22$$

$$P(1) = m + n - 6 + 22 - 12 = 0$$

$$\therefore m + n = -4 \quad \text{--- (1)}$$

$$P'(1) = 4m + 3n - 12 + 22 = 0$$

$$4m + 3n = -10 \quad \text{--- (2)}$$

$$\underline{\underline{n = -6 \quad m = 2}}$$

$$b) y = mx + 4$$

$$y = x^2 - 4x + 5$$

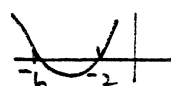
$$x^2 - 4x + 5 = mx + 4$$

$$x^2 - 4x - mx + 1 = 0$$

$$x^2 - x(4+m) + 1 = 0$$

2 pts intersection $\therefore$ 2 distinct solutions

$$\Delta > 0$$



$$(4+m)^2 - 4 > 0$$

$$m^2 + 8m + 12 > 0$$

$$(m+6)(m+2) > 0$$

$$\therefore m > -2 \text{ and } m < -6$$

c) $A(3, 2) \quad B(-1, 1)$
 $-3:2$

$$x: \frac{(2 \times 3) + (-3 \times -1)}{-3 + 2} = -9$$

$$y: \frac{(2 \times 2) + (-3 \times 1)}{-3 + 2} = -1$$

$$\underline{P(-9, -1)}$$

Question 5

a) i) BP: $y = mx + 3$

OP: $y = 2mx$

ii) $mx + 3 = 2mx$

$$3 = mx$$

$$x = \frac{3}{m}$$

$$\therefore P\left(\frac{3}{m}, 6\right)$$

iii) line $y = 6$

b) i) $y = x^{-1}$

$$\frac{dy}{dx} = -x^{-2} = -\frac{1}{x^2}$$

$$m_T = -\frac{1}{c^2} \text{ at } A\left(c, \frac{1}{c}\right)$$

$$\therefore \text{eqn tang: } y - \frac{1}{c} = -\frac{1}{c^2}(x - c)$$

ii) $B(2c, 0)$

$$C\left(0, \frac{2}{c}\right)$$

iii) $\Delta BOC = \frac{1}{2} \times \frac{2}{c} \times 2c$

$$= 2 \text{ units} \therefore \text{constant}$$

Question 6

a) i) $y = \frac{x^2}{16} \quad P(8p, 4p^2)$

$$\frac{dy}{dx} = \frac{2x}{16} = \frac{x}{8}$$

$\therefore$ at $P(8p, 4p^2) \quad m_T = p$

$\therefore$ eqn tang $y - 4p^2 = p(x - 8p)$

$$\underline{y = px - 4p^2}$$

ii) sub $T(-2, -2)$ into tangent

$$-2 = -2p - 4p^2$$

$$4p^2 + 2p - 2 = 0$$

$$(2p - 1)(p + 1) = 0 \quad \therefore p = \frac{1}{2}, -1$$

$\therefore$ Co-ord $P(4, 1) \quad p = \frac{1}{2}$

Co-ord $Z(-8, 4) \quad p = -1$

iii) $m_{RQ} = \frac{4r^2 - 4q^2}{8r - 8q} = \frac{4(r - q)(r + q)}{8(r - q)}$

$$= \frac{r + q}{2}$$

$\therefore$ eqn RQ: $y - 4r^2 = \frac{r + q}{2}(x - 8r)$

$$y - 4r^2 = \left(\frac{r + q}{2}\right)x - \frac{8r(r + q)}{2}$$

$$y - 4r^2 = \left(\frac{r + q}{2}\right)x - 4r^2 - 4rq$$

$$\underline{y = \left(\frac{r + q}{2}\right)x - 4rq} *$$

iv) If RQ is a focal chord passes through $(0, 4)$ sub into *

$$4 = -4rq$$

$$\therefore \underline{\underline{rq = -1}}$$

b) i) use sine rule in $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\underline{\underline{a \sin B = b \sin A}} \text{ ——— ①}$$

ii) In $\triangle ACD$ $\cos A = \frac{AD}{b}$
 $AD = b \cos A$

In $\triangle DBC$ $\cos B = \frac{DB}{a}$
 $DB = a \cos B$

since $c = AD + DB$

$$\therefore \underline{\underline{c = b \cos A + a \cos B}}$$

iii)

$$c^2 = (b \cos A + a \cos B)^2$$

$$c^2 = b^2 \cos^2 A + 2ab \cos A \cos B + a^2 \cos^2 B$$

$$4ab \cos A \cos B = b^2 \cos^2 A + 2ab \cos A \cos B + a^2 \cos^2 B$$

$$0 = b^2 \cos^2 A - 2ab \cos A \cos B + a^2 \cos^2 B$$

$$0 = (b \cos A - a \cos B)^2$$

$$b \cos A = a \cos B \text{ ——— ②}$$

$$\div \text{ ① by ② } \frac{a \sin B}{a \cos B} = \frac{b \sin A}{b \cos A}$$

$$\tan B = \tan A$$

$$\therefore A = B$$

$\therefore$ from ① or ②

$$a \sin B = b \sin B$$

$$\therefore \underline{\underline{a = b}}$$